

Multi-Asset Portfolio Construction Project

From Raw Stock Universe to Black–Litterman Optimal Portfolio

1. Starting Point: A Deliberately Diversified Universe

The project began with a deliberately broad and diversified universe of 138 securities, spanning multiple sectors and instruments:

- **Technology – Semiconductors & Equipment:** AMD, AVGO, QCOM, TXN, AMAT, KLAC, LRCX, MU, ADI, MCHP, MPWR, ON, SWKS
- **Technology – Software & Cloud:** MSFT, CRM, NOW, ADBE, INTU, CRWD, PANW, SNPS, CDNS, WDAY, DDOG, ORCL, FTNT
- **Technology – IT Services & Hardware:** IBM, CSCO, AKAM, NTAP, AAPL, DELL, HPE, GLW, FLEX, WDC, STX, SMC
- **Financials – Investment Banks & Asset Management:** GS, MS, BLK, BX, KKR, APO, SCHW, TROW, BK, AMP
- **Financials – Exchanges & Market Infrastructure:** CME, ICE, CBOE
- **Financials – Payments & Processing:** MA, V, PYPL, FIS, FI
- **Financials – Consumer Finance & Services:** COF, SYF, IBKR, VRSK, MSCI
- **Consumer Discretionary – E-Commerce & Retail:** AMZN, HD, LOW, TJX, ROST, ULTA, AZO, ORLY, BBY
- **Consumer Discretionary – Restaurants:** MCD, CMG, SBUX, YUM, DPZ
- **Consumer Discretionary – Apparel & Footwear:** NKE, DECK, RL, CROX
- **Consumer Discretionary – Hotels & Gaming:** MAR, HLT, MGM, LVS
- **Consumer Discretionary – Automotive:** TSLA, GM
- **Industrials – Aerospace & Defense:** GE, TDG
- **Industrials – Construction & Building:** NVR, DHI, LEN
- **Industrials – Diversified Industrials & Equipment:** ROP, TDY, CPRT, HUBB, MSI, POOL
- **Communication Services:** EBAY, LYV
- **Healthcare – Pharma & Biotech:** LLY, ABT, TMO, AMGN, PHAR, NRIX, ELV, CHGCY, TAK
- **Healthcare – Funds & ETFs:** ETIHX, FBIOX, FBTDX, VHCIX, PRHSX, SHSKX, PJP, IXJ, XBI, FXH
- **Utilities:** NEE, DUK, D, AEP, SO, ED, EIX, EXC, XEL, AES, CNP, ETR, EVRG, LNT, ATO, NI, NRG, AWK, WTRG, PEG, PPL

This starting universe intentionally mixes **individual stocks, ETFs**, and sector/thematic funds across regions and business models to capture a rich cross-section of risk premia.

The rest of the project is essentially:

“How do we go from a big, noisy, overlapping set of names to a stable, diversified optimal portfolio using modern portfolio theory and Bayesian enhancements (Black–Litterman)?”

2. Script 1 – Data Preparation and Baseline Risk Metrics

Objective: Turn raw price data into clean return series and foundational risk measures.

2.1 From prices to log returns

Using project_data.xlsx, Script 1:

- Loaded monthly adjusted prices for all 138 tickers.
- Converted dates to proper Date objects and sorted them chronologically.
- Cast the data into an xts time series object for time-series operations.

Then it computed **log returns** for every asset i :

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right)$$

The first NA row was dropped, and the log returns were saved as log_returns_df.rds for reuse in later steps.

2.2 Excess returns over risk-free

To put everything on a risk-adjusted footing:

- Assumed a monthly risk-free rate $r_f = 0.0003$.
- Constructed excess returns r

$$r_{i,t}^{excess} = r_{i,t} - r_f$$

and saved them as `excess_returns_df.rds`.

2.3 Variance-covariance and correlation matrices

From the matrix of log returns, Script 1 estimated:

- The variance-covariance matrix $\Sigma = \text{cov}(\mathbf{R})$, saved as `var_cov_matrix.rds` and `var_cov_matrix_full.csv`.
- The correlation matrix $\rho = \text{cor}(\mathbf{R})$, saved as `cor_matrix.rds` and `correlation_full.csv`.

These provide the first look at how volatile each asset is and how strongly assets move together—key inputs into portfolio construction.

3. Script 2 – Narrowing the Universe via Correlation Filtering

Objective: Reduce redundancy and improve numerical stability by removing strongly collinear assets.

3.1 Measuring correlation structure

Script 2 started from `log_returns_df.rds` and `cor_matrix.rds`, and computed summary statistics of the off-diagonal correlations (average, min, max), providing a sense of how crowded the universe is in terms of co-movement.

3.2 Graph-based filtering: independent set of low-overlap assets

To systematically reduce redundancy, the script:

- Defined a **correlation threshold** $|\rho_{\{ij\}}| > 0.5$.
- Built an **adjacency matrix** where two assets were connected if their absolute correlation exceeded 0.5.
- Interpreted this adjacency matrix as a graph using *igraph*.
- Ran a *greedy independent set* algorithm that selected a subset of tickers such that no two selected assets had $|\rho| > 0.5$.

The result was a filtered list of tickers (*tickers_filtered.csv* → *tickers_filtered_final.csv*) that preserved sector and style diversity but **reduced correlation** clustering.

3.3 Cleaned returns and filtered Σ

From the filtered tickers, Script 2:

- Subsetted the returns matrix.
- Removed any columns with NA or zero variance.
- Computed a **filtered correlation matrix** (*correlation_filtered.csv*).
- Computed a filtered sample covariance matrix *sigma_sample* (*sigma_sample_full.csv* and *sigma_sample_lower.csv*).

3.4 Eigenvalue diagnostics

To check numerical stability, the script computed the eigenvalues of *sigma_sample* and the **condition number** $\kappa = \lambda_{\max} / \lambda_{\min}$. This motivated exploring **regularized covariance estimators** in later scripts.

4. Script 3 – Single Index Model (SIM)

Objective: Use a factor model to generate a structured covariance matrix that reduces noise and estimation error.

4.1 Equal-weighted “market” factor

From the filtered returns, Script 3 defined an equal-weighted market return:

$$R_{m,t} = \frac{1}{N} \sum_{i=1}^N R_{i,t}$$

4.2 Regressions for alpha, beta, and idiosyncratic risk

For each asset i , the script estimated a **Single Index Model**:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t}$$

It extracted:

- α_i : intercept (stock-specific return component),
- β_i : sensitivity to the market factor,
- $\sigma^2_{\epsilon_i}$: idiosyncratic variance (variance of residuals).

4.3 SIM variance–covariance matrix

Using the SIM structure, covariance between assets is:

- For $i \neq j$:

$$\text{Cov}(R_i, R_j) = \beta_i \beta_j \sigma_m^2$$

- For $i = j$:

$$\text{Var}(R_i) = \beta_i^2 \sigma_m^2 + \sigma_{\epsilon_i}^2$$

where $\sigma_m^2 = \text{Var}(\mathbf{R}_m)$ is the variance of the market factor.

Script 3 built the SIM covariance matrix `sigma_SIM` and exported full and lower-triangular versions.

$$\Sigma_{\text{SIM}} = \beta \beta^\top \sigma_m^2 + \text{diag}(\sigma_{\epsilon}^2)$$

4.4 Visual diagnostics

Heatmaps of `sigma_SIM` showed how **factor risk** (via β) generates broad blocks of covariance, while **idiosyncratic risk** shows up on the diagonal. SIM provides a **more structured, lower-dimensional representation** of risk than the raw sample covariance.

5. Script 4 – Constant Correlation Model

Objective: Construct a regularized covariance matrix under the assumption that **all off-diagonal correlations are equal** to the average correlation.

From `correlation_filtered.csv`, we computed the average pairwise correlation (approximately 0.2125).

Script 4 then:

- Built a **constant correlation matrix** with 1s on the diagonal and the average correlation on all off-diagonal entries.
- Combined this with sample standard deviations σ_i from `sigma_sample_full.csv` to form a **constant-correlation covariance matrix**, where D is the diagonal matrix of standard deviations.

$$\Sigma_{\text{const}} = D C_{\text{const}} D$$

This model enforces a simple, robust structure: assets differ mainly by their volatilities, and they share a common average level of co-movement. It reduces sampling noise compared to the full sample covariance, especially in high dimensions.

6. Script 5 – Shrinkage Covariance Matrix (Excel-style)

Objective: Mimic the shrinkage method used in the Excel workbook by **scaling down off-diagonal** covariances to temper extreme co-movement estimates.

From the filtered sample covariance matrix `sigma_sample`, Script 5 implemented:

- **Same variances** on the diagonal:

$$\Sigma_{ii}^{\text{shrink}} = \Sigma_{ii}$$

- **Scaled covariances** off the diagonal by a factor $\lambda = 0.6$:

$$\Sigma_{ij}^{\text{shrink}} = 0.6 \cdot \Sigma_{ij}, \quad i \neq j$$

This corresponds to a **linear shrinkage** toward a diagonal matrix:

$$\Sigma^{\text{shrink}} = (1 - \gamma)\Sigma + \gamma D$$

with $\gamma = 0.4$ and D being the diagonal covariance (variances only).

Intuition:

- We trust individual **volatility estimates** reasonably well.
- We are **less confident in the precise pairwise covariances**, so we pull them toward zero.
- This improves out-of-sample stability and mitigates over-fitting.

The result is `sigma_shrinkage_full.csv` – a more stable, regularized covariance estimator.

7. Script 6 – Markowitz Efficient Frontier and Optimal Portfolios

Objective: Use the various covariance estimators (sample, SIM, constant-correlation, shrinkage) to construct the Markowitz efficient frontier and identify optimal portfolios under each risk model.

7.1 Mean returns and sample Σ

From `log_returns_df.rds` and the filtered universe, Script 6 estimated mean log returns μ_i and used the filtered sample covariance `sigma_sample` as the baseline Σ .

7.2 Efficient frontier without short-sale constraints

Using standard mean–variance theory, the script computed:

- Defined:

$$\Sigma^{-1}, \quad \mathbf{1}, \quad \mu$$

- Computed Markowitz scalars:

” Ask ChatGPT $B = \mathbf{1}^\top \Sigma^{-1} \mu, \quad C = \mu^\top \Sigma^{-1} \mu, \quad D = AC - B^2$

- For a grid of target returns μ^* , solved for portfolio weights w that satisfy:

- $\mathbf{1}^\top w = 1,$
- $\mu^\top w = \mu^*,$

and minimized variance via Lagrange multipliers.

and minimized variance via Lagrange multipliers, yielding an efficient frontier table (`efficient_frontier.csv`) with σ and expected return for each portfolio.

7.3 Individual asset risk–return points

Script 6 also computed each ticker's expected return μ_i and volatility $\sigma_i = \sqrt{\Sigma_{ii}}$, allowing individual assets to be plotted against the efficient frontier.

7.4 Tangency portfolios under different Σ

Given a risk-free rate r_f , for each covariance estimator Σ (sample, constant-correlation, SIM, shrinkage), Script 6 computed a tangency portfolio:

$$w^{\text{tang}} \propto \Sigma^{-1}(\mu - r_f \mathbf{1})$$

normalized so weights sum to 1. This produced:

- **Sample Σ tangency portfolio,**
- **SIM tangency portfolio,**
- **Constant-correlation tangency portfolio,**
- **Shrinkage tangency portfolio.**

Weights and summaries were exported as `optimal_portfolios_weights.csv` and `optimal_portfolios_summary.csv`. Plots showed the efficient frontier (blue), individual assets (red), and optimal portfolios (triangles) for visual comparison.

8. Black–Litterman Extension – Benchmark-Anchored Optimal Portfolio

Objective: Move from a pure math-driven optimal portfolio to one that blends a reasonable benchmark and subjective views using the Black–Litterman framework.

8.1 Constructing a benchmark portfolio

From `optimal_portfolios_weights.csv` (weights from Sample, ConstantCorr, SIM, Shrinkage), we built a benchmark by averaging weights for each ticker across all four tangency portfolios:

$$w_i^{\text{bench}} = \frac{1}{4} \sum_{\text{methods}} w_{i,\text{method}}$$

Weights were normalized to sum to 1 and saved as `benchmark_weights.csv`. This consensus benchmark reflects a blend of different risk models and the common exposures they favor.

8.2 Implied equilibrium returns with normalizing factor

Using Σ (often the shrinkage version) and w_{bench} , Black–Litterman infers equilibrium

returns that make the benchmark optimal:

1. Compute benchmark variance:

$$\text{Var}(R_{\text{bench}}) = w_{\text{bench}}^{\top} \Sigma w_{\text{bench}}$$

2. Compute benchmark expected return (from historical data) and excess return over risk-free:

$$\mathbb{E}[R_{\text{bench}}], \quad \mathbb{E}[R_{\text{bench}}] - r_f$$

3. Derive the **normalizing factor** δ (risk-aversion parameter):

$$\delta = \frac{\mathbb{E}[R_{\text{bench}}] - r_f}{\text{Var}(R_{\text{bench}})}$$

4. Equilibrium (implied) **excess returns**:

$$\pi_{\text{excess}} = \delta \Sigma w_{\text{bench}}$$

5. Equilibrium **total returns**:

$$\mu_{\text{equil}} = r_f + \pi_{\text{excess}}$$

These equilibrium returns (`bl_equilibrium_returns.csv`) serve as the market-consistent prior in Black–Litterman.

8.3 Encoding investor views (P, Q, Ω)

Black–Litterman adds views:

- **P**: $K \times N$ matrix encoding which assets each view applies to.
- **Q**: $K \times 1$ vector of view excess returns.
- **Ω** : $K \times K$ covariance matrix of view uncertainty.

In this project, an illustrative view was set on the asset with the highest historical sample mean, asserting its future excess return aligns with its sample excess return. This view was encoded in P, Q, and Ω (saved as `bl_views_P.csv`, `bl_views_Q.csv`, `bl_views_Omega.csv`).

8.4 Black–Litterman posterior returns and covariance

Combining the equilibrium prior with views yields posterior excess returns and covariance:

- Posterior excess returns:

$$\mu_{\text{excess}}^{BL} = [(\tau \Sigma)^{-1} + P^{\top} \Omega^{-1} P]^{-1} [(\tau \Sigma)^{-1} \pi_{\text{excess}} + P^{\top} \Omega^{-1} Q]$$

- Posterior covariance:

$$\Sigma^{BL} = \Sigma + [(\tau \Sigma)^{-1} + P^{\top} \Omega^{-1} P]^{-1}$$

Posterior returns $\mu_{BL_total} = r_f + \mu_{BL_excess}$ balance the benchmark-implied equilibrium with investor views.

8.5 BL optimal portfolio and benchmark comparison

Using μ_{BL_total} and Σ_{BL} , the script computed a BL tangency portfolio:

$$w^{BL} \propto (\Sigma^{BL})^{-1} (\mu^{BL} - r_f \mathbf{1})$$

Weights were normalized and exported along with benchmark weights and portfolio summaries (`bl_weights_benchmark_and_optimal.csv`, `bl_portfolio_summary.csv`).

8.6 BL efficient frontier and visualization

With μ_{BL} and Σ_{BL} , the script built a BL efficient frontier (*bl_efficient_frontier.csv*) and plotted:

- BL frontier,
- BL asset points,
- Benchmark portfolio,
- BL optimal portfolio.

This showed where the benchmark lies relative to the BL frontier and how the BL optimal portfolio improves the risk–return trade-off while remaining anchored to the benchmark and views.

9. Conceptual Takeaways

1. Structured reduction of the investable universe

Starting from 138 names, correlation filtering produced a more numerically stable, less redundant subset of securities while preserving sector breadth.

2. Competing covariance models to manage estimation error

We contrasted sample covariance, the Single Index Model, constant-correlation, and shrinkage covariance. This highlights the trade-off between flexibility and robustness in risk modeling.

3. Efficient frontier and tangency portfolios as a research canvas

The Markowitz frontier and tangency portfolios under different Σ show how diversification dominates individual securities and how risk model choice affects optimal portfolios.

4. Black–Litterman ties it back to a benchmark and investor views

Rather than blindly trusting historical means or a purely mathematical optimum, BL uses the benchmark as a prior and incorporates explicit views with quantified confidence.

5. From theory to implementation

Scripts 1–6 operationalize core concepts in portfolio theory: mean–variance optimization, covariance estimation, factor models, shrinkage, efficient frontiers, and Bayesian updating. The final output is not just a portfolio, but a framework that can be updated as data and views evolve.